Implantation and Testing of WFMA Stimulators in Macaque*

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***Abstract*— Our on-going work to develop an intracortical visual prosthesis has motivated the design, fabrication, and testing of a Wireless Floating Microelectrode Array (WFMA) stimulator. This implantable device can be used for an electrical stimulation interface in the peripheral and the central nervous system. Previously, its use in a sciatic nerve rodent model was described. Here implantation of two WFMAs in motor cortex of two NHP (macaque – two devices/animal) is presented. Preliminary functional tests show the implanted devices to be fully functional with stimulation-induced motor movements obtained. Functional testing is on-going.**

I. Introduction

There are many reports of intracortical electrode array systems having limited implanted lifetimes substantially due to issues related to the reliability of percutaneous connector systems [1,2]. For cochlear implants and other experimental functional neural stimulators some progress has been reported for overcoming the limitations of connector-based systems through the use of implantable devices that are powered by, and communicated with, wireless means, [3]. However, these approaches do not fully eliminate the tethering cable that connects to the electrode structures. For a chronic intracortical system, eliminating the tethering cable is generally regarded as highly desirable, perhaps essential, in order to allow the electrode arrays to "float" with the brain. Deployment of a fully integrated wireless intracortical array has been largely conceptual, and reports in the literature have been limited to in-vitro demonstrations, acute non-survival animal studies, or hybrid demonstrations that utilize some form of primary power other than a wireless source [4][5]. The design of fully wireless systems that integrate electronics with electrode arrays have notable challenges related to packaging and biocompatibility (physical and material) with the cortex [6]. As part of our Intracortical Visual Prosthesis (ICVP) development, we have developed a wireless intracortical stimulating electrode device intended for long-term cortical implantation in humans. The WFMA is a wireless intracortical stimulation device that provides constant-current charge-balanced electrode driving, and the sending of reverse telemetry of electrode voltage waveforms for sixteen activated iridium oxide film (AIROF) electrodes. We have previously reported the use of the WFMA for chronic implantation in rat sciatic nerve. [7,8,9], and these implanted devices have remained functional for approximately 16 months, until animal euthanasia. Here, we report the implantation and preliminary functional testing of the WFMA in motor cortex of Macaque. These surgical experiences in non-human primate, have particular significance for a planned clinical trial of the ICVP in humans.

II. Methods

A. WFMA Description

The physical structure of the WFMA is shown in Fig 1, and the function has been described in [7,8,9]. Each WFMA contains 16 AIROF electrodes combined with wireless electronics that allows for powering and communication over a transcutaneous inductive link operating at 5MHz.

WFMA base structures that contained the electrodes within the ceramic superstructure base were fabricated by Microprobe for Life Science (MLS) and sent to the Illinois Institute of Technology (IIT). At IIT, the WFMA base structure was completed into a WFMA by the addition of the application-specific-integrated circuit (ASIC), the power/communication coil, and the polymeric encapsulation. Following WFMA encapsulation curing, the electrodes were activated via the wireless link as described in [10].

Testing of WFMA mechanical samples, using a high-speed insertion tool [11] was done to refine the tool electrical driving parameters so that the insertion speed reached 1-2 m/sec. Consistent with the design of the insertion tool, the WFMA devices were loaded into the tool collets and the WFMAs with the tool were sterilized by Sterrad. This procedure was repeated for implantations in two Rhesus monkeys (macaque), with surgical procedures separated about 2 weeks apart.

B. Surgical Description

The surgical implantations followed an approved animal protocol at the University of Chicago, with some procedural differences between the two surgeries being related to other hardware implanted at the time of WFMA implantations. The other hardware supports auxiliary testing with the animals not related to the WFMA testing.

For both surgical procedures: After sedation and intubation, the animal was placed in prone position, and the head was fixed in a parallel bar stereotaxic frame. The hair on the head was clipped, and the cranial vault area above the ears was prepped with iodine-based solution. A longitudinal

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incision of the scalp was made along the midline, and the scalp spread with sutures. The likely location of the motor strip was accomplished by measurement of the distance from orbits and occiput based on imaging, and by skull landmarks such as the coronal suture. A craniotomy was performed measuring approximately 2.5cm by 2cm. After hemostasis, a durotomy was performed with a scalpel and microscissors. The leaflets of the dura were retracted. Surface cortical stimulation was done with hand-held electrodes while observing the animal's hand and face, resulting in the identification of the motor strip, specifically relative to the face and hand motor areas. To insert the WFMAs into the cortex, the electromechanical insertion tool was attached to a 6-axis manipulator connected to the stereotactic frame. After adjusting the manipulator so that the tip of the insertion tool was appropriately located, and perpendicular to the brain, the collect containing the WFMA was loaded into the tool, and electric drive to the tool motor was triggered. This procedure was repeated for a second WFMA device. After implantation, we approximated the dural leaflets, and covered with a layer of gelfoam.

III. Results

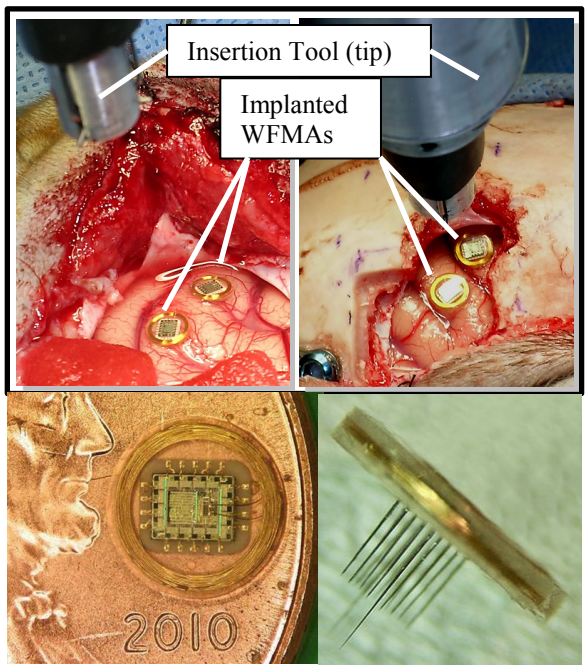

Figure 1. – Top: Photograph of two WFMA stimulators implanted in motor cortex of two macaques. Note the tip of the inserter tool above the implanted WFMAs, after withdrawal, immediately following insertion. Bottom: WFMA shown next to US penny and in side view. Dimensions: 5mm diam x 0.5mm thickness

Fig 1 shows the WFMA devices implanted into the brain of the two animals. The WFMAs implantations were successfully accomplished as evidenced by the visual observation of no surface blood vessel leakage and stable post-insertion positioning. Immediately following the surgery, after scalp closure, the electrical operation of the WFMAs was confirmed by verifying command and reverse telemetry communication with the implanted devices.

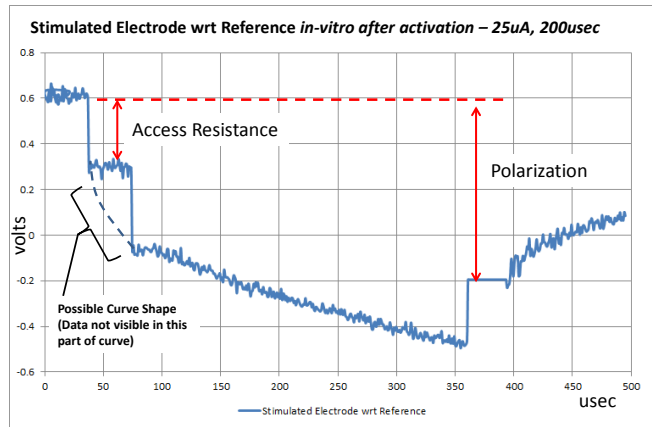

Figure 2. – Representative electrode voltage transient for constant current pulse recorded in vitro.

A representative electrode voltage transient waveshape, in vitro, as transmitted over the WFMA wireless reverse telemetry link is shown in Fig 2. Here can been seen the portions of the waveform relevant to the determination of two critical parameters for assessing the stimulation capacity of electrodes: Access Resistance, and Polarization. A specialized feature magnification capability of the WFMA system is specifically designed to enhance the determination of these parameters.

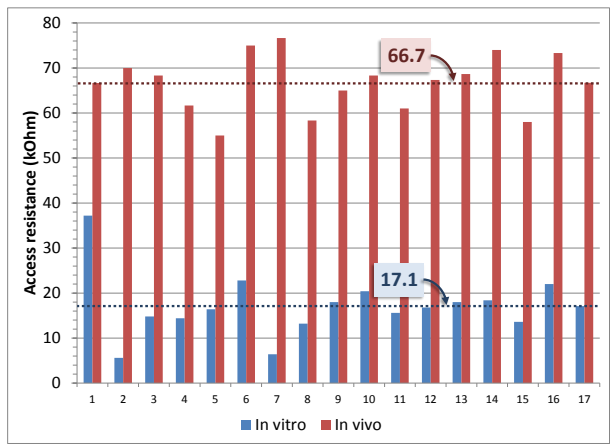

Figure 3. – Comparison of Access Resistance for Array #1 in vitro and in vivo. Average values for both groups indicated by dashed lines and histogram #17.

Functional testing of the WFMAs, in both animals is on-going, with preliminary results available at the time of this writing. Initial testing of the first animal, one week post implant, showed continued electrical operation and visual evidence of hand and face movements caused by the wireless commanding of the WFMAs. Due to anesthesia time limits in this first testing session, more extensive testing in the awake animal is scheduled for the near future. Initial testing while under anesthesia in the second animal showed electrical operation of the WFMAs as evidenced by the reverse telemetry. Voltage transients, via reverse telemetry, for 30uA of stimulation current were recorded for all 32 electrodes.

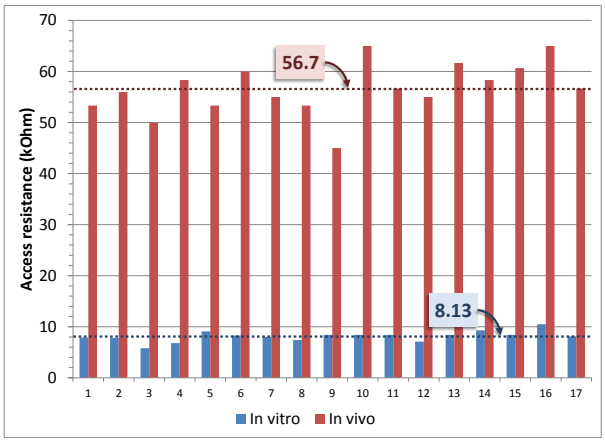

Figure 4. – Comparison of Access Resistance for Array #2 in vitro and in vivo. Average values for both groups indicated by dashed lines and histogram #17.

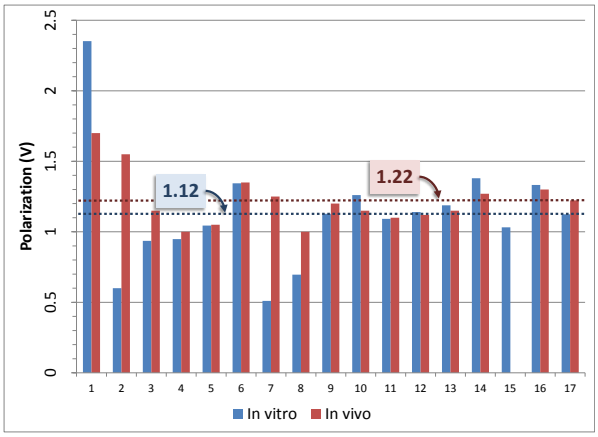

Figure 5. – Comparison of Polarization for Array #1 in vitro and in vivo. Average values for both groups indicated by dashed lines and histogram #17.

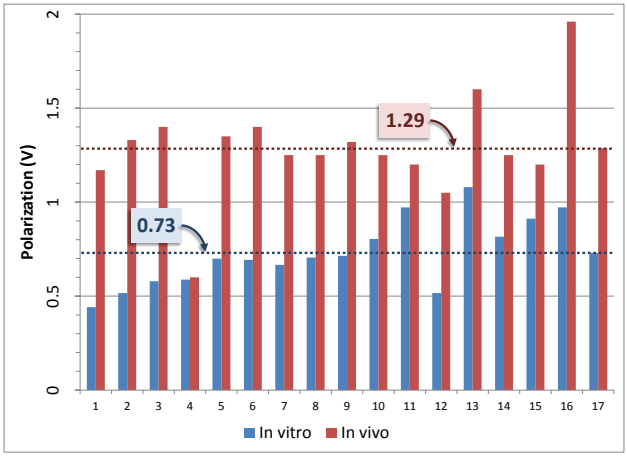

Figure 2. – Comparison of Polarization for Array #2 in vitro and in vivo. Average values for both groups indicated by dashed lines and histogram #17.

For all 32 electrodes (each WFMA has 16 electrodes), in animal 2, voltage transient waveforms were recorded, in vivo, two weeks after implantation. For each electrode, the access resistance and polarization were extracted from the voltage transient and compared with those same parameters as recorded in vitro. These data are shown in Figs 3, 4, 5, 6. For each chart, a 17th histogram value is shown that is the averages, also indicated by the horizontal dashed lines.

Functional testing, to observe motor responses, of both animals is on-going, albeit with some challenges coordinating the primate training with the visual observations of the stimulation induced motor activity and the configuration of extracorporeal telemetry controller units.

IV. Discussion

The reverse telemetry with the transmission of diagnostic information about the operation of the WFMAs and the electrode voltage transient waveforms is extremely helpful for knowing that the implanted devices are electrically functional and that they are delivering current through the electrodes. As long as the electrode polarization is not excessive (i.e. exceeding the water window), the programmed stimulation current is being delivered in an electrochemically "safe" manner. It is interesting to note that when comparing the in vitro and in vivo access resistances, there is a significant increase for the in vivo values over the in vitro ones. Yet, there is only a minor change of the polarization. This suggest that perhaps a thinner AIROF film could provide for a lower overall voltage transient magnitude since it is known that a portion of the access resistance is contributed by thicker films. More study of this is needed. For the in-vivo transient shown in Fig 3, most of the voltage drop is associated with access resistance rather than polarization. This suggests that thinner AIROF films might be used to reduce the overall magnitude of the transient. Paradoxically, an electrode that has lower charge capacity in-vitro may have an overall lower total voltage excursion when used in-vivo if the magnitude of the voltage transient were to be less. This issue requires further study which we are pursuing in a cortical rodent model.

V. Conclusion

Implantation of WFMAs in cortex using high-speed insertion seems feasible and may form the basis for studies with larger numbers of devices. We expect that the chronic functionality of these cortical WFMAs may follow that of the peripheral rodent WFMA studies, and months of experimental results may be obtained. These first-ever demonstrations of WFMA cortical implantation are significant in that a fully implantable 16-channel intracortical stimulator requiring no percutaneous connection has numerous uses for neural modulation within motor and sensory cortex.

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